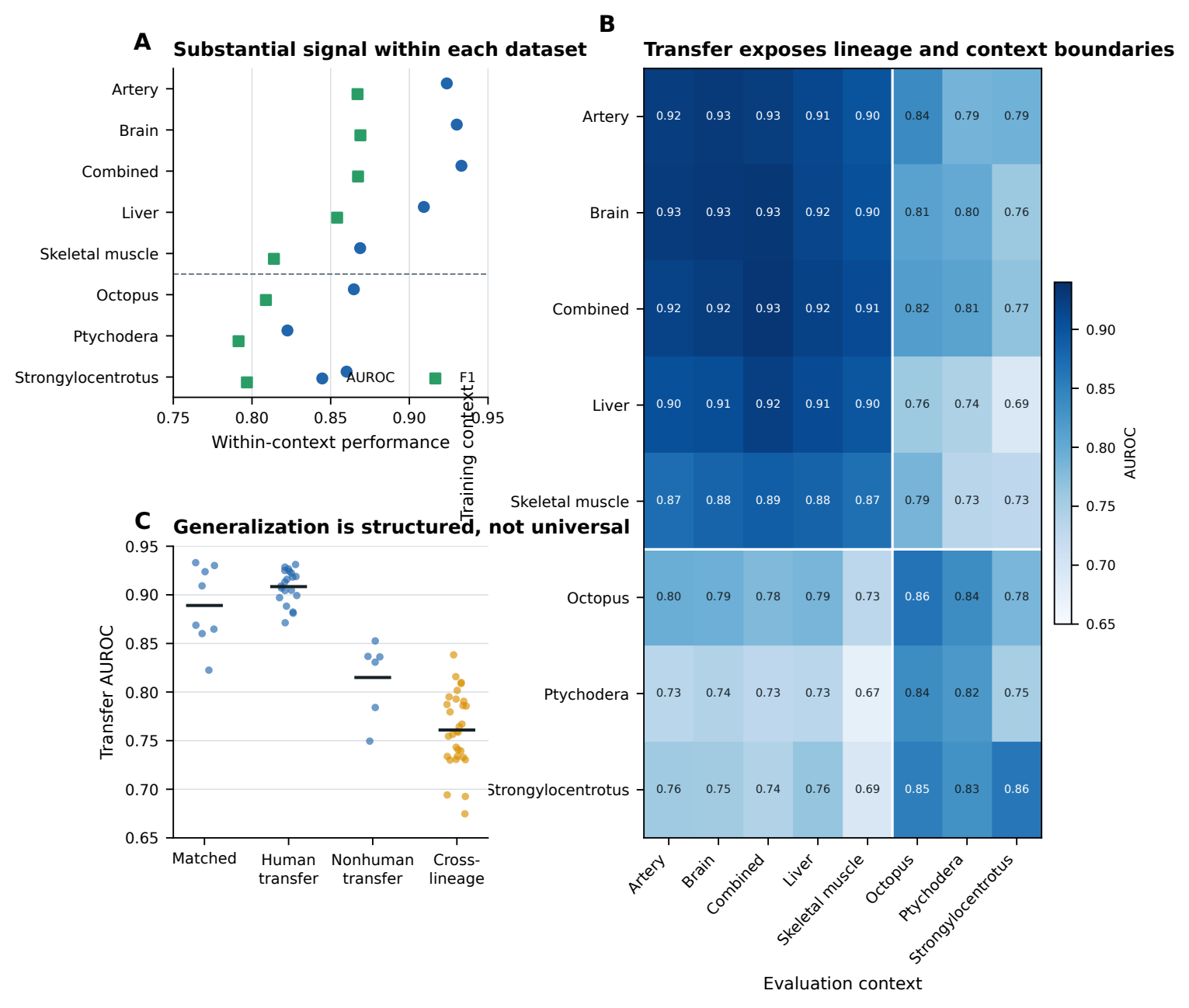

A
Substantial signal within each dataset
Artery
Brain
Combined
Liver
Skeletal muscle
Octopus
Ptychodera
Strongylocentrotus
AUROC
F1
0.75
0.80
0.85
0.90
0.95
Within-context performance
Training context
B
Transfer exposes lineage and context boundaries
Artery 0.92 0.93 0.93 0.91 0.90 0.84 0.79 0.79
Brain 0.93 0.93 0.93 0.92 0.90 0.81 0.80 0.76
Combined 0.92 0.92 0.93 0.92 0.91 0.82 0.81 0.77
Liver 0.90 0.91 0.92 0.91 0.90 0.76 0.74 0.69
Skeletal muscle 0.87 0.88 0.89 0.88 0.87 0.79 0.73 0.73
Octopus 0.80 0.79 0.78 0.79 0.73 0.86 0.84 0.78
Ptychodera 0.73 0.74 0.73 0.73 0.67 0.84 0.82 0.75
Strongylocentrotus 0.76 0.75 0.74 0.76 0.69 0.85 0.83 0.86
Artery
Brain
Combined
Liver
Skeletal muscle
Octopus
Ptychodera
Strongylocentrotus
Evaluation context
AUROC
0.90
0.85
0.80
0.75
0.70
0.65
C
Generalization is structured, not universal
Transfer AUROC
0.95
0.90
0.85
0.80
0.75
0.70
0.65
Matched
Human transfer
Nonhuman transfer
Cross-lineage